AI-Driven Screening for Carbon Capture Materials

Predicting CO_2 Uptake in Metal-Organic Frameworks via Machine Learning

Aashima Chemical Engineering & Machine Learning GitHub Repository

Domain: Mass Transfer & Adsorption Theme: Materials Discovery Task: Regression

Abstract. Metal-Organic Frameworks (MOFs) are nanoporous crystalline materials with extraordinary internal surface areas — a single gram can exceed the surface area of a football field. These structural properties make them promising candidates for selectively adsorbing CO_2 from industrial flue gas streams, a critical step in Carbon Capture and Storage (CCS) technologies. Over 90,000 MOF structures have been synthesized to date, yet experimentally measuring CO_2 uptake capacity remains prohibitively slow. This work presents a complete machine learning pipeline that predicts CO_2 uptake at flue gas conditions (0.15 bar, 298 K) purely from geometric structural descriptors across a dataset of 324,426 MOFs. Four models are trained and compared — LightGBM, XGBoost, Random Forest, and a Feed-Forward Neural Network. LightGBM achieves the best test R^2 of **0.9266**, demonstrating that structural geometry alone is sufficient for accurate large-scale screening of carbon capture candidates.

1. Introduction

Carbon capture and storage (CCS) is one of the most critical technological pathways for reducing atmospheric CO_2 levels and mitigating climate change. Industrial flue gas streams — emitted from power plants and manufacturing facilities — are composed primarily of nitrogen (∼85%) and CO_2 (∼15%) at near-atmospheric pressures. Selectively separating and capturing CO_2 from this mixture is the central challenge.

Metal-Organic Frameworks (MOFs) are crystalline porous materials self-assembled from inorganic metal nodes and organic linker molecules. Their extraordinary structural tunability — adjustable pore size, geometry, surface chemistry, and topology — makes them among the most promising solid-state sorbents for selective CO_2 uptake. The internal surface area of a MOF can exceed 6,000 m^2/g, providing vast adsorption capacity within a small mass.

Despite their promise, identifying the best-performing MOF from the enormous chemical space of synthesizable structures remains a grand challenge. Over 90,000 MOF structures have been reported, and the space of hypothetical MOFs extends to millions. Experimental characterization of each candidate is time-prohibitive. Computational screening using Grand Canonical Monte Carlo (GCMC) simulation accelerates this process but remains computationally expensive at the scale of hundreds of thousands of structures.

Machine learning offers a transformative alternative. If a model can learn the mapping from a MOF's geometric descriptors to its CO_2 uptake, predictions become nearly instantaneous. This project builds and evaluates such a model using four ML approaches, with a focus on honest evaluation — predictions are made purely from structural geometry, not from other simulation outputs.

2. Dataset

The dataset used in this work is a large-scale computational MOF screening database containing **324,426 MOF structures**. Each structure is characterized by 42 columns spanning geometric descriptors, simulation outputs at multiple pressure and temperature conditions, and categorical structural identifiers.

2.1. Target Variable

The prediction target is `CO2_uptake_P0.15bar_T298K` in mmol/g, representing CO_2 uptake at standard flue gas conditions — 0.15 bar partial pressure and 298 K temperature. The raw distribution is strongly right-skewed, with a mean of 0.52 mmol/g and a maximum of approximately 6 mmol/g, motivating a logarithmic transformation prior to training.

2.2. Structural Input Features

Only geometric and structural descriptors are retained as model inputs. Simulation-derived outputs are deliberately excluded to ensure honest predictive screening capability:

- `surface_area` — total internal surface area [m^2/g]
- `void_fraction` — fraction of unit cell that is empty pore space
- `void_volume` — total accessible pore volume

[cm^3/g]

- `volume` — unit cell volume [$Å^3$]
- `weight` — molecular weight of the unit cell [u]
- `largest_free_sphere_diameter` — largest sphere that can pass through the pore network [Å]
- `largest_included_sphere_diameter` — largest sphere that fits within the pore [Å]
- `largest_included_sphere_along_free_sphere_path` [Å]
- `topology` — framework connectivity type (11 unique values)
- `functional_groups` — chemical functional groups (∼400 unique)
- `metal_linker` — metal node identity (7 unique values)
- `organic_linker1`, `organic_linker2` — organic linker identities (58 unique each)

3. Methodology

3.1. Data Cleaning

The raw dataset contains several categories of columns that must be removed before modelling to avoid noise and data leakage.

Error and uncertainty columns (10 columns): These record the statistical uncertainty in GCMC simulation results — essentially how well the simulation converged. They carry no structural information about the MOF and would only add noise to the model. They were identified automatically using the keyword `error` in column names and dropped entirely.

Leakage columns (17 columns): These are simulation outputs at other pressure/temperature conditions, including working capacity (vacuum swing, temperature swing), CO_2/N_2 selectivity, binary uptake values, and CO_2 uptake at 0.10 bar/363 K and 0.70 bar/413 K. Including any of these creates severe data leakage — the model learns trivial cross-condition correlations such as "high uptake at 0.10 bar implies high uptake at 0.15 bar" instead of learning genuine structure-to-property relationships. A model trained with leakage columns appears to achieve $R^2 > 0.99$ but is useless for screening because it requires the very simulations it is meant to replace.

After removing the `MOFname` string identifier and dropping approximately 2,000 rows with missing values (<1% of the dataset), the final clean dataset contains **322,000+** samples and **32 features**.

3.2. Categorical Encoding

The dataset contains four categorical columns requiring numeric encoding before model training. Different strategies were applied based on cardinality:

Column	Encoding	Unique
`functional_groups`	Frequency encoding	∼400
`topology`	One-hot encoding	11
`metal_linker`	One-hot encoding	7
`organic_linker1/2`	Integer (as-is)	58

Table 1: Encoding strategy per categorical column.

Frequency encoding replaces each functional group label with the number of times it appears in the dataset. This avoids the 400-column explosion that one-hot encoding would create, while providing the model with a meaningful signal about how common each functional group is.

One-hot encoding creates a separate binary column for each category. Used for topology and metal linker because both have few values and no meaningful numeric ordering — treating them as integers would impose a false ordinal ranking on unordered categories.

After encoding, all boolean columns created by one-hot encoding were converted to integer (0/1) for LightGBM compatibility.

3.3. Feature Engineering

Three physically motivated interaction features were derived from the existing geometric descriptors. These capture structural relationships that individual features alone cannot express:

$$f_1 = \frac{\text{surface_area}}{\text{volume} + \epsilon} \quad \text{(surface density)} \tag{1}$$

$$f_2 = \frac{\text{void_fraction}}{\text{sphere_diameter} + \epsilon} \quad \text{(pore density)} \tag{2}$$

$$f_3 = \text{void_volume} \times \text{surface_area} \quad \text{(capacity proxy)} \tag{3}$$

where $\epsilon = 10^{-9}$ prevents division by zero.

3.4. Train / Validation / Test Split

The dataset is partitioned into three non-overlapping sets using random seed 42 for reproducibility:

Split	Fraction	Rows
Train	70%	∼227,000
Validation	15%	∼48,600
Test	15%	∼48,600

Table 2: Dataset split sizes.

The validation set is used exclusively for early stopping during training. The test set is held out completely until final evaluation, ensuring that reported metrics reflect true out-of-sample performance.

3.5. Feature Scaling

`StandardScaler` is applied to normalize all features to zero mean and unit variance. The scaler is fit exclusively on the training set and then applied to the validation and test sets using training statistics. This prevents leakage of test-set distributional information into the training process.

Although tree-based models (LightGBM, XGBoost, Random Forest) are theoretically invariant to monotonic feature scaling, standardization is applied consistently across all four models for fair comparison and to ensure stable training of the FFN.

3.6. Target Transformation

The CO_2 uptake target is transformed before training:

$$\tilde{y} = \ln(1 + y) \tag{4}$$

This `log1p` transformation compresses the right-skewed distribution, making it approximately symmetric and improving the model's ability to fit the full range of uptake values without being dominated by outliers. Predictions are reversed using the inverse transformation:

$$\hat{y} = e^{\tilde{y}} - 1 \tag{5}$$

All reported metrics (R^2, MSE, MAE) are computed on the original mmol/g scale after back-transformation.

4. Models

4.1. LightGBM

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework developed by Microsoft that builds an ensemble of decision trees sequentially, where each new tree corrects the residual errors of the previous ones.

Its key innovation is **leaf-wise tree growth**: instead of growing trees level by level (as in most other boosting methods), LightGBM selects the single leaf with the highest loss reduction at each step. This produces deeper, more asymmetric trees that achieve higher accuracy with fewer iterations. It also uses **histogram binning** — feature values are bucketed into discrete bins before splitting, dramatically reducing memory usage and training time on large datasets like our 324,426-row MOF database.

Key hyperparameters used in this work:

- `n_estimators = 3000`, `learning_rate = 0.02` — many small steps for careful, gradual learning
- `num_leaves = 100`, `max_depth = 10` — moderate tree complexity to balance expressiveness and overfitting
- `min_child_samples = 50` — requires at least 50 samples per leaf, preventing overly specific splits
- `subsample = colsample_bytree = 0.7` — each tree uses 70% of rows and features, adding diversity
- `reg_alpha = reg_lambda = 0.1` — L1 and L2 regularization
- Early stopping with 50 rounds patience on validation loss

4.2. XGBoost

XGBoost (Extreme Gradient Boosting) is also a gradient boosting framework but uses **level-wise tree growth**, expanding all leaves at the same depth before moving deeper. This makes trees more regular and symmetric, which tends to generalize better on some datasets at the cost of slightly lower raw accuracy.

XGBoost incorporates second-order gradient information (both first and second derivatives of the loss function) in its tree-building process, making each split decision more mathematically precise. It also has built-in L1 (`reg_alpha`) and L2 (`reg_lambda`) regularization directly in the objective function.

In this work, XGBoost achieves the **smallest train-test gap** (0.004) of all four models, demonstrating excellent generalization to unseen MOF structures — a critical property for real screening applications.

4.3. Random Forest

Random Forest is a **bagging ensemble** of decision trees, where each tree is trained independently (not sequentially) on a bootstrap sample of the training data with a random subset of features considered at each split. Final predictions are the average across all trees.

This combination of data randomness (bootstrap sampling) and feature randomness (random feature subsets) produces highly decorrelated trees, and their averaged prediction has much lower variance than any individual tree. Unlike boosting, Random Forest trees are grown to full depth with no pruning, but variance is controlled through averaging.

In our experiments, Random Forest exhibits the largest test MSE among the four models. The smoothing effect of averaging across 500 trees systematically underestimates CO_2 uptake for high-performance MOFs — precisely the candidates most relevant for carbon capture applications.

4.4. Feed-Forward Neural Network (FFN / MLP)

The FFN is a three-layer Multi-Layer Perceptron with the following architecture:

- **Input layer**: 32 features (after encoding and engineering)
- **Hidden layer 1**: 256 neurons, ReLU activation, Batch Normalization, Dropout ($p = 0.3$)
- **Hidden layer 2**: 128 neurons, ReLU activation, Batch Normalization, Dropout ($p = 0.3$)
- **Hidden layer 3**: 64 neurons, ReLU activation
- **Output layer**: 1 neuron (regression output)

Trained using the Adam optimizer with learning rate 10^{-3} and Mean Squared Error loss. StandardScaler is *critical* for the FFN — without it, features on vastly different scales (e.g. surface area in thousands vs. void fraction between 0 and 1) cause gradient instability and extremely slow convergence. The FFN achieves competitive performance at test $R^2 = 0.916$ without any of the tree-specific structural inductive biases present in the other models.

5. Results

Model	Train R^2	Test R^2	MSE	MAE
LightGBM	**0.9542**	**0.9266**	**0.0211**	**0.0813**
FFN / MLP	0.9340	0.9160	0.0230	—
XGBoost	0.9135	0.9095	—	0.1485
Random Forest	0.9481	0.9042	0.0270	0.0940

Table 3: Performance on held-out test set. All metrics on original mmol/g scale after `expm1` back-transform.

5.1. Comparative Analysis

LightGBM achieves the highest test R^2 (0.9266) and lowest MSE (0.0211), confirming that its leaf-wise growth strategy and early stopping effectively balance model complexity with generalization. The train-test gap of 0.028 indicates mild overfitting that could be further reduced with Optuna-based hyperparameter search.

XGBoost demonstrates the most consistent train-test behaviour (gap of only 0.004), making it the most reliable model for deployment on truly novel MOF structures not represented in the training distribution.

FFN performs remarkably close to the tree-based models (0.916) despite making no assumptions about feature interactions. This suggests the relationships between geometric descriptors and CO_2 uptake are smooth enough for neural approximation.

Random Forest underperforms on the test set relative to its training R^2. The averaging mechanism, while reducing variance, introduces systematic bias on the high-uptake tail of the distribution — the region of greatest practical interest for screening.

5.2. LightGBM Diagnostic Plots

The following diagnostic visualizations were generated for LightGBM (the best-performing model) after training:

Predicted vs. Actual plot: A scatter plot of predicted CO_2 uptake against actual values on the test set. Points lying on the diagonal red line represent perfect predictions. The plot shows tight clustering along the diagonal for uptake values between 0 and 3 mmol/g, with slight dispersion at higher values, consistent with the right-skewed nature of the target distribution.

Residuals plot: Residuals (actual − predicted) plotted against predicted values. The residuals are approximately centered around zero across the full prediction range, indicating no systematic bias. A slight funnel pattern at higher predicted values is observed, reflecting the model's reduced accuracy for the rarest high-uptake MOFs.

Residuals distribution: A histogram of residuals approximating a symmetric bell curve centered at zero, confirming well-calibrated predictions without directional bias.

Feature importance (gain): Bar chart of LightGBM split-gain importance showing `surface_area`, `void_fraction`, and `void_x_sa` as dominant predictors — physically consistent with adsorption theory.

6. SHAP Analysis

SHAP (SHapley Additive exPlanations) is a game-theoretic framework for interpreting machine learning model predictions at both the global and local (per-sample) level. Unlike standard feature importance metrics that report aggregate split statistics, SHAP values quantify the exact contribution of each feature to each individual prediction.

For a prediction $\hat{f}(x)$, the SHAP decomposition is:

$$\hat{f}(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{6}$$

where ϕ_0 is the base rate (mean prediction over the training set), ϕ_i is the SHAP value for feature i (its contribution to the deviation from the base rate), and M is the total number of features.

6.1. Global Feature Importance via SHAP

SHAP summary plots rank features by their mean absolute SHAP value across the test set. For the LightGBM model, the top features by SHAP importance are:

- `surface_area` — the single most impactful feature; higher surface area consistently drives positive SHAP values (increased predicted uptake), physically reflecting greater available adsorption sites
- `void_fraction` — pore space fraction is strongly positively correlated with uptake; MOFs with higher void fraction have more space for CO_2 molecules to occupy
- `void_x_sa` — the engineered interaction feature capturing combined surface area and pore volume effect ranks highly, validating the physical motivation for its creation
- `largest_included_sphere_diameter` — pore size determines whether CO_2 molecules (kinetic diameter ∼3.3 Å) can access the pore network; negative SHAP values emerge for very small pores that exclude CO_2
- `functional_groups` (frequency-encoded) — certain functional group families are consistently associated with higher or lower uptake, suggesting chemical affinity effects beyond purely geometric properties

6.2. Local Explanations

SHAP waterfall plots for individual MOF predictions reveal how each feature pushes the prediction above or below the dataset mean. For high-uptake MOFs, large positive SHAP contributions from `surface_area` and `void_fraction` combine to produce predictions well above the base rate. For low-uptake MOFs, small pore diameters and low void fractions contribute negatively.

6.3. SHAP Interaction Effects

SHAP dependence plots reveal nonlinear interactions between features. Notably, the relationship between `surface_area` and predicted uptake is strongly modulated by `void_fraction`: high surface area combined with low void fraction (dense but highly textured MOFs) yields lower uptake than the same surface area with high void fraction. This interaction is captured by the engineered `void_x_sa` feature and is consistent with known adsorption physics.

7. Discussion

7.1. Physical Validation

The convergence of SHAP importance, engineered feature performance, and correlation analysis on `surface_area` and `void_fraction` as top predictors is physically meaningful. In adsorption theory, CO_2 uptake is directly proportional to the accessible surface area and pore volume of the sorbent. The model has learned this physically correct relationship from data alone, without any domain constraints.

7.2. Honest Screening Philosophy

The deliberate exclusion of all simulation-derived features is the most important design decision in this work. A model trained with leakage columns (selectivity, working capacity, uptake at other conditions) trivially achieves $R^2 > 0.99$ but cannot be deployed for actual screening: it requires GCMC simulation results as inputs, which defeats the purpose of building a screening model in the first place.

The R^2 values reported here (0.9266 for LightGBM) represent genuine structure-to-property prediction from geometry alone — a true computational screening capability.

7.3. Limitations and Future Work

The current geometric feature set, while effective, does not capture atomic-level chemistry. Incorporating metal electronegativity, linker aromaticity indices, or charge distribution descriptors could push R^2 above 0.95 without introducing leakage. Additionally:

- **Optuna hyperparameter tuning** could close the LightGBM train-test gap from 0.028 to below 0.01
- **Graph Neural Networks** operating directly on the MOF crystal graph could capture topological information not accessible to tabular models

- **Stacking ensemble** combining LightGBM, XGBoost, and FFN predictions could further improve test R^2
- **Active learning** could identify the most informative MOFs to simulate next, guiding experimental discovery

8. Conclusion

This project demonstrates that geometric structural descriptors alone are sufficient to predict CO_2 uptake across 324,426 MOFs with a test R^2 of 0.9266 using LightGBM. The complete pipeline — from leakage-free feature selection and categorical encoding through log-transform preprocessing, StandardScaler normalization, boosted tree training, and SHAP-based interpretability — provides a practical and physically grounded template for rapid in-silico screening of novel MOF candidates.

Among the four models evaluated, LightGBM achieves the best absolute performance, XGBoost achieves the best generalization (smallest train-test gap), and the FFN demonstrates the viability of neural approaches for this task. The SHAP analysis confirms that the learned feature importances are physically meaningful, with surface area and void fraction dominating predictions in alignment with established adsorption theory.

The methodology developed here can be extended to other target properties (N_2 uptake, selectivity, working capacity) and other material classes (zeolites, covalent organic frameworks), establishing a reusable framework for data-driven materials discovery in carbon capture and beyond.

References

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Source code available at github.com/aashima2310/carbon_capture_project